

ROS 2

Sensor sampling and image processing

Roberto Masocco

`roberto.masocco@uniroma2.it`

University of Rome Tor Vergata

Department of Civil Engineering and Computer Science Engineering

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TOR VERGATA
UNIVERSITÀ DEGLI STUDI DI ROMA

Recap

ROS 2 offers a common framework for the development of **robotics software**, providing services for

- **organizing** and **building** a **distributed** software architecture;
- establishing mostly self-configured **inter-process communication** among modules;
- modules **configuration** and **management**;
- process **launch** configuration.

This lecture is [here](#).

For the following code examples, a host Linux installation is required.

Roadmap

- 1 Sensor sampling
- 2 Image processing
- 3 Software tools of the trade
- 4 Examples: a target detection pipeline

Roadmap

- 1 Sensor sampling
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Sensor sampling basics

From theory to practice

Sampling a sensor consists of reading **measurements** from it, to be fed to a control loop or some other subsystem.

It requires:

- the definition of a **sampling frequency**;
- the implementation of an **encoding**;
- the application of **post-processing** steps (e.g., **filtering**).

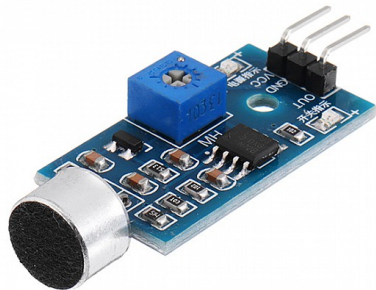


Figure 1: Analog sound sensor.

Sensor sampling with ROS 2

Driver modules

Sampling is generally handled by **microcontrollers**, but when the sampling frequency is *not too high*, e.g., periods down to some ms, it may be carried out by a higher-level device.

To implement a ROS 2 sensor sampling module, one has to develop a **driver node**, i.e., an application that

- configures the **sensor hardware** to run as required;
- ensures **stable sampling frequency** and **low jitter**;
- outputs data with a **standard interface** and **low latency**.

The achievement of the first goal depends on the **sensor**, the second on the **system** (hardware and software!), while the third one is solved by ROS 2 (**messages**, **QoS**).

Must take the best of both worlds: robotics and system programming!

Anatomy of a driver node

Guidelines and best practices

In essence, a **driver node** always consists of

- an **enable service**, to be called to start or stop the sampling;
- a **hardware configuration** routine, to be run at startup or when enabled;
- a **sampling loop**, to be run at a fixed frequency in a separate **thread**;
- a **publisher** using a common message type and an appropriate QoS policy;
- a set of **parameters** to configure the sensor and the sampling loop;
- **launch files** and **configuration files**, to configure remapping rules and node behaviour.

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Vision in robotics

Sensors characteristics

Visual sensors, commonly referred to as **cameras**, are sensors that provide **images** as output, encoded in **frames**, *i.e.*, **matrices** of data points. They are usually made of

- one or more **optical lenses**, to focus light on the sensor;
- a **sensor**, to convert light into electrical signals;
- a **processing unit**, to convert electrical signals into images, optionally applying **post-processing** steps.



Figure 2: RGB USB camera.

Vision in robotics

Sensors characteristics

Light might not belong to the visible band of the spectrum.



Figure 3: Intel RealSense D435i depth camera: RGB and IR sensors.

Vision in robotics

Sensors characteristics

Cameras are usually characterized by

- **resolution**, *i.e.*, the number of pixels in the image;
- **field of view**, *i.e.*, the angular extension of the scene (horizontal and vertical);
- **frame rate**, *i.e.*, the number of frames per second;
- **dynamic range**, *i.e.*, the ratio between the maximum and minimum measurable light intensity.



Figure 4: Intel RealSense T265 tracking camera: two fisheye sensors.

Vision in robotics

Sensors characteristics

Cameras, and image processing algorithms in general, are usually characterized by a trade-off between resolution and frame rate.



Figure 5: ZED Mini stereo tracking camera.

Vision in robotics

Main use cases

Visual sensors are usually employed in robotics for

- **object detection**, *i.e.*, identifying objects in the scene;
- **object tracking**, *i.e.*, following objects in the scene;
- **localization**, *i.e.*, estimating the robot's position in the environment;
- **mapping**, *i.e.*, building a model of the environment.

Using the sensor is not enough: **algorithms** are needed to perform these tasks.
Such algorithms usually run in separate modules.

Types of frames

RGB frame



Figure 6: RGB frame: each pixel contains at least the intensities of the red, green and blue components of the corresponding point in the scene.

Types of frames

IR frame



Figure 7: IR frame: each pixel contains the intensity of the corresponding point in the scene.

Types of frames

Depth map frame



Figure 8: Depth map frame: each pixel contains the distance of the corresponding point from the camera.

Lens distortion

Camera calibration and rectification

Pinhole cameras generally introduce **radial distortion** in the images they produce, *i.e.*, **straight lines appear curved**. This is due to how the light enters the camera through the lens.

A **camera calibration** procedure is needed to estimate the **intrinsic camera parameters** of the **distortion model**, so that a **rectification map** can then be applied to each frame.



Figure 9: Checkerboard pattern used for camera calibration in the presence of radial distortion.

Lens distortion

Camera calibration and rectification

The calibration procedure usually involves moving a **checkerboard** of known dimensions in front of the camera, and taking several pictures of it from different angles.

An **estimation model** can then be used to estimate the parameters of the distortion model, which can then be used to build the rectification map.



Figure 10: A frame from a checkerboard calibration procedure.

Lens distortion

Camera calibration and rectification

The **rectification map** is a **lookup table** that associates each pixel in the original frame with a pixel in the rectified frame, **interpolating** when necessary.

It is usually stored in **configuration files**, and must be applied to each frame before it can be used by other algorithms.

ROS 2 offers the [cameracalibrator](#) tool in the `camera_calibration` package to **perform camera calibration.**



Figure 11: Checkerboard pattern after rectification.

Event cameras

A new frontier in computer vision for robotics

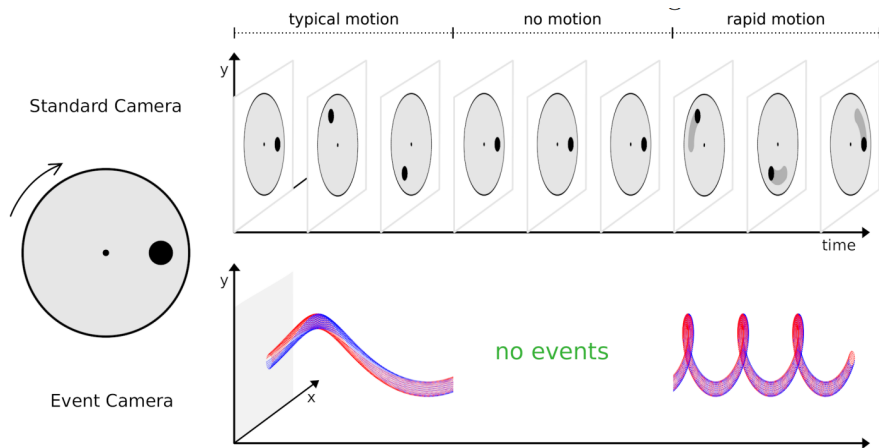


Figure 12: Working principle of event cameras.

Event cameras

A new frontier in computer vision for robotics

Also called **neuromorphic cameras**, they are a new type of visual sensor that provides **events** instead of **frames**, and are **complementary** to traditional cameras.

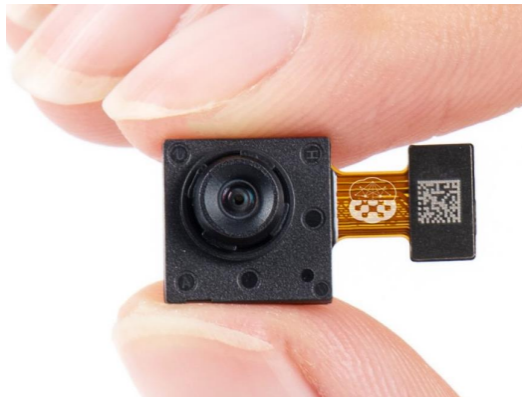


Figure 13: Event camera from SynSense.

Event cameras

A new frontier in computer vision for robotics

They are characterized by

- **no motion blur**;
- **high dynamic range**;
- **very low latency** (microseconds);
- **low power consumption** (milliwatts).

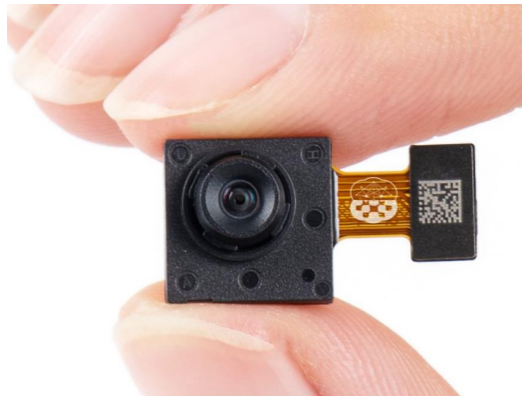


Figure 13: Event camera from SynSense.

Event cameras

A new frontier in computer vision for robotics

They are very recent (circa 2008), and are still in the **research phase**. They are being evaluated in **high-speed** and **high-dynamic range** applications, such as

- **autonomous driving;**
- **localization and mapping;**
- **object tracking;**
- **industrial automation, inspection, and visual servoing;**
- **artificial intelligence.**

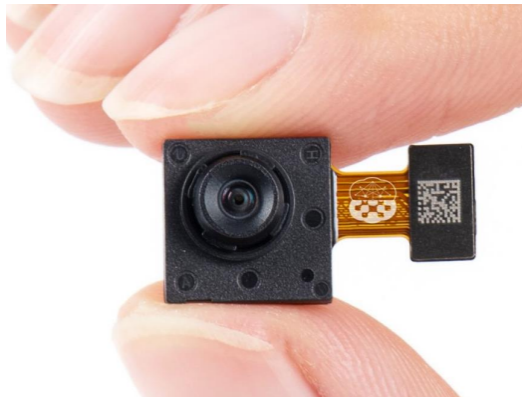


Figure 13: Event camera from SynSense.

Event cameras

A new frontier in computer vision for robotics

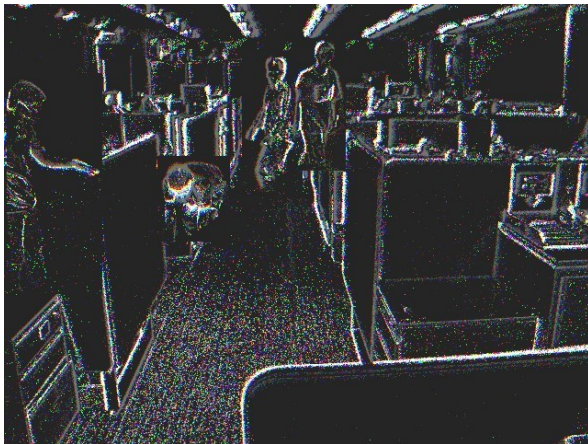


Figure 14: Frame from an event camera.

Roadmap

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OpenCV

The de-facto standard library for computer vision

OpenCV is the largest open-source library for real-time **computer vision** and **image processing**. It has been developed initially by Intel, and evolved in a **cross-platform** library supporting **C++** and **Python**.

It ships with hundreds of **APIs** and **algorithms** to perform a wide variety of image processing and computer vision tasks and computations.

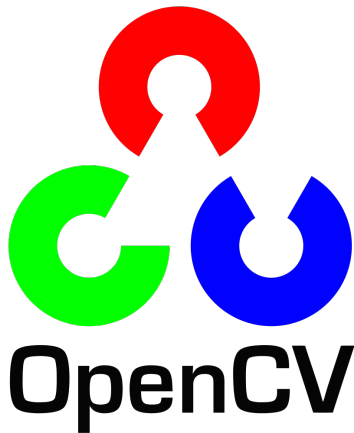


Figure 15: OpenCV logo.

OpenCV

The de-facto standard library for computer vision

In Linux, it is available as **binary packages**, but the full range of features can be enabled only by **configuring a source build**.

Its **elementary data type** is a **matrix**: `cv::Mat`.

It is **heavily optimized** to run on:

- **parallel CPUs**;
- **GPUs** (e.g., `cv::cuda`, `cv::ocl`);
- **embedded devices** (e.g., Nvidia Jetson).

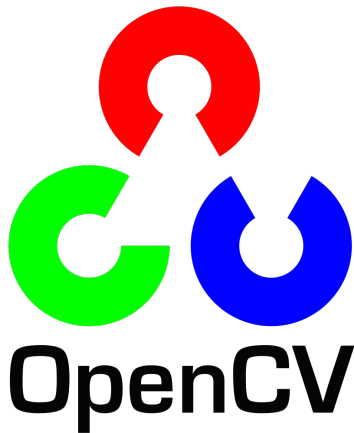


Figure 15: OpenCV logo.

ROS tools for image processing

A quick overview

ROS 2 provides a wide range of **tools** to acquire, process, and transmit images.

The most important ones are

- `sensor_msgs/Image`: the standard ROS **message type** to transmit **images** over the DDS layer;
- `sensor_msgs/CameraInfo`: the standard ROS **message type** to transmit and parse **camera calibration parameters**;
- [image_transport](#): a ROS package to **transmit images and metadata** over the RMW layer efficiently;
- [camera_info_manager](#): a library to parse, use, and store **camera calibration parameters** from configuration files and messages inside ROS nodes.

ROS tools for image processing

The Image message

```
1 std_msgs/Header header
2
3 uint32 height
4 uint32 width
5
6 # This may be "RGB", "BGR", "RGBA", "BGRA", "YUV", "Bayer" ...
7 string encoding
8
9 uint8 is_bigendian
10 uint32 step
11 uint8[] data
```

Listing 1: Definition of the sensor_msgs/msg/Image message.

ROS tools for image processing

The CameraInfo message

```
1 std_msgs/Header header
2
3 uint32 height
4 uint32 width
5
6 string distortion_model
7 float64[] d # Distortion coefficients
8 float64[9] k # Intrinsic raw camera matrix
9 float64[9] r # Rectification matrix (rotated stereo cameras)
10 float64[12] p # Projection camera matrix
11
12 uint32 binning_x
13 uint32 binning_y
14 RegionOfInterest roi
```

Listing 2: Definition of the sensor_msgs/msg/CameraInfo message; go have a look at it.

ROS tools for image processing

Sending images over the DDS

Using images and video streams in general over the DDS layer is **not trivial**, since

- ① we would like some kind of **specialized transport strategy** to optimize throughput and latency;
- ② the DDS specification is **not optimized** to transmit **large** and **variable-size** data chunks over potentially **lossy** networks, e.g., WiFi.

`image_transport` is a ROS 2 library that provides **wrappers** for `Image+CameraInfo` **topic publishers** and **subscribers**, allowing to transmit images using different **transports**:

- `raw`: the default transport, which uses the `sensor_msgs/Image` message;
- `compressed`: uses the `sensor_msgs/CompressedImage` message, automatically converting frames to JPEG or PNG format;
- whatever you want to implement as a **plugin**.

ROS tools for image processing

A note on topics and their statistics

Beware!

When measuring image topics publishing rates with `ros2 topic hz`, especially **over a lossy network**, you **may** get **high variance** or straight out **wrong results**.

This is due to:

- the **Python DDS implementation**;
- the **DDS struggling with large messages**;
- the `hz` Python tool **buffering** messages into a window-based filter.

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ArUco detection pipeline

A real example from LDC22

Examples for this lecture are in the [cpp/image_processing](#) directory.

There are **three nodes**:

- ① `ros2_usb_camera`: **acquires images** from a USB camera;
- ② `aruco_detector`: **detects ArUco fiducial markers** in a video stream;
- ③ `rqt_image_view`: forked version of the official ROS 2 **video stream visualizer**.

These three nodes form a **pipeline** to **detect ArUco markers** in a video stream, and show them in a **GUI**. They are **completely configurable**, and **optimized** to run on **GPU** and similar hardware. They make use of **all the features we have seen so far** including composition, benefitting of **zero-copy** data transfer and shared address space.

Multiple instances of these nodes were active during the Leonardo Drone Contest 2022!